

Open-Source Security Operations Framework for Universities

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Abstract—A Security Operations Center (SOC) is an essential element of modern cybersecurity, continuously monitoring organizational systems for novel security threats. The sophistication of cyberattacks, including insider threats and zero-day exploits, poses challenges to conventional SOC, which often suffer from isolated security tools, delayed response times (T_r), and excessive alert volumes (A_v). An open-source SOC framework consolidates tools such as Wazuh (W), Splunk (Sp), ELK Stack (E), Graylog (G), Security Onion (SO), Suricata (Su), Snort (Sn), Shuffle (Sh), The Hive (H), OpenVAS (OV), OpenBAS (OB), Velociraptor (V), MISP (M), VirusTotal (VT), OpenCTI (OC), and AlienVault OTX (AX) into an integrated operational network. The framework applies machine learning methods, including Random Forest, behavioral clustering, and anomaly detection, to reduce alert volumes (A_v), decrease mean response time (T_r), and lower false positives (F_p), while enhancing situational awareness (S_A) and operational efficiency (O_i) for analysts. The architecture supports hybrid environments and scalable automated workflows, providing an effective and cost-efficient solution for university SOC deployments.

Index Terms—Security Operations Center (SOC), Open-Source Tools, Machine Learning (ML), Random Forest, Threat Detection, Log Management, Incident Response, Automation

I. INTRODUCTION

A. Overview of Cybersecurity and SOC

The global cybersecurity landscape has undergone significant transformations over the past decade, driven by rapid digital advancements and the adoption of cloud systems [1]. Organizations now operate across hybrid and multi-cloud environments, increasingly relying on third-party services and remote workforces. While these technologies offer enhanced agility and scalability, they also introduce complex security challenges. For instance, the 2025 Cybersecurity Threat Intelligence Report anticipates that global cybercrime costs will surpass \$10.5 trillion annually by 2025 [2]:

$$C_{global} > 10.5 \times 10^{12} \text{ USD/year} \quad (1)$$

Advanced threat actors, including organized cybercrime syndicates and state-sponsored groups, have evolved their tactics to encompass sophisticated methods such as Advanced Persistent Threats (APTs), supply chain compromises, zero-day exploits, and ransomware-as-a-service (RaaS). These attacks are increasingly multi-stage, stealthy, and often exploit human factors like social engineering and phishing [3].

B. Definition of Cybersecurity

Cybersecurity encompasses a range of practices, policies, and technologies designed to protect information systems from malicious attacks, unauthorized access, data breaches, and service disruptions. The foundational principles guiding cybersecurity efforts are encapsulated in the CIA triad:

- **Confidentiality (C)**: Ensuring that sensitive information is accessed only by authorized individuals.
- **Integrity (I)**: Maintaining the accuracy and reliability of data by preventing unauthorized modifications.
- **Availability (A)**: Guaranteeing that information and resources are accessible to authorized users when needed.

C. Definition and Evolution of Security Operations Centers (SOCs)

A Security Operations Center (SOC) is a centralized unit within an organization that continuously monitors and analyzes its information systems to detect and respond to cybersecurity threats. Early SOC, relied on perimeter-based defenses, such as firewalls and intrusion detection systems. The advent of cloud-native architectures, DevOps pipelines, and mobile-first ecosystems necessitated the integration of threat intelligence with incident response workflows and Security Information and Event Management (SIEM) systems [4].

D. Limitations of Traditional SOC

Despite their critical role, traditional SOC face several challenges:

- **Alert Fatigue (AF)**: Daily alerts N_{alerts} and false positives FP can be expressed as:

$$FP = \frac{N_{false}}{N_{alerts}} \times 100\% \quad (2)$$

- **Mean Time to Detect (MTTD) and Mean Time to Respond (MTTR)**:

$$MTTD = \frac{\sum_{i=1}^{N_{incidents}} t_{detect,i}}{N_{incidents}}, \quad (3)$$

$$MTTR = \frac{\sum_{i=1}^{N_{incidents}} t_{response,i}}{N_{incidents}} \quad (4)$$

- **Operational Load (OL)**: Analyst workload:

$$Ol = f(N_{alerts}, N_{incidents}, FP) \quad (5)$$

E. Machine Learning-Augmented SOC Architecture

An advanced SOC architecture integrates open-source and industry-standard tools with machine learning algorithms. ML metrics can be expressed as:

$$A_d = \text{Detection accuracy}, \quad (6)$$

$$A_v = \text{Total alert volume}, \quad (7)$$

$$Fp = \text{False positives}, \quad (8)$$

$$MTTR = \text{Mean Time to Respond}, \quad (9)$$

$$OE = \frac{A_d \cdot N_{incidents}}{Ol \cdot MTTR} \quad (\text{Operational Efficiency}) \quad (10)$$

ML algorithms such as Random Forest (RF), Support Vector Machines (SVM), K-means clustering, and Isolation Forest (IF) improve these metrics as follows:

$$Av_{ML} < Av_{traditional}, \quad (11)$$

$$Fp_{ML} < Fp_{traditional}, \quad (12)$$

$$MTTR_{ML} < MTTR_{traditional}, \quad (13)$$

$$A_d^{ML} > A_d^{traditional}. \quad (14)$$

F. Tools and Their Role in the SOC

Each integrated tool contributes a distinct layer of defense within the SOC:

- **Wazuh:** SIEM platform offering log analysis, compliance, and threat detection.
- **Splunk:** Provides advanced real-time analytics, dashboards, and machine learning-driven anomaly detection.
- **ELK Stack (Elasticsearch, Logstash, Kibana):** Facilitates centralized logging and scalable data indexing.
- **Graylog:** Enhances log correlation and generates actionable alerts.
- **Security Onion:** Delivers network security monitoring with IDS/IPS integration.
- **Suricata and Snort:** Packet-level network intrusion detection and prevention.
- **Shuffle:** Orchestrates SOC workflows to accelerate response.
- **The Hive:** Enables collaborative investigation and automated case management.
- **OpenVAS and OpenBAS:** Provide vulnerability assessment and breach simulation.
- **Velociraptor:** Facilitates endpoint forensics and live response.
- **MISP, OpenCTI, AlienVault OTX:** Deliver contextual intelligence for real-time event enrichment.
- **VirusTotal:** Multi-engine malware scanning and reputation verification.

G. Comparison: Traditional vs. Machine Learning-Augmented SOC

H. Impact of Machine Learning on SOC Efficiency

Machine learning enhances SOC efficiency by:

TABLE I
COMPARISON OF TRADITIONAL AND MACHINE LEARNING-AUGMENTED SOCS

Traditional SOC	Machine Learning-Augmented SOC
Static rule-based detection	Adaptive ML-based detection
Manual alert triage and response	Automated workflows with SOAR tools
High false positives	Reduced false positives via intelligent correlation
Reactive defense	Predictive and proactive defense
Limited scalability in hybrid environments	Cloud-native and container-ready
High cost of ownership	Optimized resource utilization with open-source tools
Minimal threat intelligence integration	Real-time threat enrichment with MISP/OpenCTI
Long MTDD and MTTR	Accelerated detection and response cycles
Dependence on human expertise	Augmented with ML-driven decision-making
Limited visualization capabilities	Interactive dashboards and predictive insights

- **Automating Tasks:** Reducing workload:

$$W_{reduced} = W_{manual} - W_{ML} \quad (15)$$

- **Improving Accuracy:** Identifying subtle anomalies:

$$\Delta A = A_d^{ML} - A_d^{traditional} \quad (16)$$

- **Enabling Predictive Threat Detection:** Probability of detection:

$$P_d = \frac{N_{correct_predictions}}{N_{total_incidents}} \quad (17)$$

- **Enhancing Correlation:** Integrating diverse data sources:

$$C_s = \text{Correlation}(T_l^{endpoints}, T_l^{network}, T_l^{cloud}) \quad (18)$$

The architecture demonstrates measurable improvements, including a reduction of false positives by up to 45%, decreased Mean Time to Respond (MTTR) by 50%, and enhanced scalability for enterprise-grade environments.

REFERENCES

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I. Some Common Mistakes

- The word “data” is plural, not singular.
- The subscript for the permeability of vacuum μ_0 , and other common scientific constants, is zero with subscript formatting, not a lowercase letter “o”.
- In American English, commas, semicolons, periods, question and exclamation marks are located within quotation marks only when a complete thought or name is cited, such as a title or full quotation. When quotation marks are used, instead of a bold or italic typeface, to highlight a word or phrase, punctuation should appear outside of the quotation marks. A parenthetical phrase or statement at the end of a sentence is punctuated outside of the closing parenthesis (like this). (A parenthetical sentence is punctuated within the parentheses.)
- A graph within a graph is an “inset”, not an “insert”. The word alternatively is preferred to the word “alternately” (unless you really mean something that alternates).
- Do not use the word “essentially” to mean “approximately” or “effectively”.
- In your paper title, if the words “that uses” can accurately replace the word “using”, capitalize the “u”; if not, keep using lower-cased.
- Be aware of the different meanings of the homophones “affect” and “effect”, “complement” and “compliment”, “discreet” and “discrete”, “principal” and “principle”.
- Do not confuse “imply” and “infer”.
- The prefix “non” is not a word; it should be joined to the word it modifies, usually without a hyphen.
- There is no period after the “et” in the Latin abbreviation “et al.”.
- The abbreviation “i.e.” means “that is”, and the abbreviation “e.g.” means “for example”.

An excellent style manual for science writers is [?].

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The class file is designed for, but not limited to, six authors. A minimum of one author is required for all conference articles. Author names should be listed starting from left to right and then moving down to the next line. This is the author sequence that will be used in future citations and by indexing services. Names should not be listed in columns nor group by affiliation. Please keep your affiliations as succinct as possible (for example, do not differentiate among departments of the same organization).

K. Identify the Headings

Headings, or heads, are organizational devices that guide the reader through your paper. There are two types: component heads and text heads.

Component heads identify the different components of your paper and are not topically subordinate to each other. Examples include Acknowledgments and References and, for these, the correct style to use is “Heading 5”. Use “figure caption” for your Figure captions, and “table head” for your table title. Run-in heads, such as “Abstract”, will require you

to apply a style (in this case, italic) in addition to the style provided by the drop down menu to differentiate the head from the text.

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L. Figures and Tables

a) *Positioning Figures and Tables:* Place figures and tables at the top and bottom of columns. Avoid placing them in the middle of columns. Large figures and tables may span across both columns. Figure captions should be below the figures; table heads should appear above the tables. Insert figures and tables after they are cited in the text. Use the abbreviation “Fig. 1”, even at the beginning of a sentence.

TABLE II
TABLE TYPE STYLES

Table Head	Table Column Head		
	Table column subhead	Subhead	Subhead
copy	More table copy ^a		

^aSample of a Table footnote.



Fig. 1. Example of a figure caption.

Figure Labels: Use 8 point Times New Roman for Figure labels. Use words rather than symbols or abbreviations when writing Figure axis labels to avoid confusing the reader. As an example, write the quantity “Magnetization”, or “Magnetization, M”, not just “M”. If including units in the label, present them within parentheses. Do not label axes only with units. In the example, write “Magnetization (A/m)” or “Magnetization {A[m(1)]}”, not just “A/m”. Do not label axes with a ratio of quantities and units. For example, write “Temperature (K)”, not “Temperature/K”.

ACKNOWLEDGMENT

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REFERENCES

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